



ORIGINAL PAPER

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Machine learning-based urban densification: extending roof ridge lines for sustainable housing extension using generative adversarial networks

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Abstract

To address the challenges of global urbanization and housing shortages, implementing practical densification approaches often necessitates tailoring solutions based on the local context and prevailing housing typologies. However, such expansion strategies have been limited to unidirectional stacking or single-direction extensions while also heavily relying on designers' previous experience and subjective judgment. Therefore, this study proposes a novel machine learning (ML)-based framework for generating multi-directional house extension options, enabling efficient and contextually appropriate residential densification. Unlike existing approaches limited to unidirectional expansion and subjective designer input, our framework automatically identifies suitable land surfaces, conducts voxel-based generation of extensions, and incorporates customizable, structurally valid prefabricated components. A case study of Walthamstow, a neighborhood in North London, UK, demonstrates the framework's potential for significant residential densification. Key findings reveal that our proposed data-driven approach can generate scalable densification solutions tailored to diverse residential building types and neighborhoods, offering a promising strategy to reduce urban sprawl, alleviate the housing crisis, and minimize environmental impact through efficient, automated, and contextually sensitive design. This ML-based framework significantly advances automated densification strategies, providing a practical tool for sustainable urban development.

Keywords Housing density, Densification, Machine learning, Generative adversarial networks (GAN), Housing extension, Sustainable urban planning

1 Introduction

In the context of global urbanization, the urban population in the UK is projected to increase from 57 million in 2021 to 63 million by 2035, with the rural population declining from 11 to 9 million (Mohajeri et al., 2023).

This dramatic increase in urban dwellers implies a significant shortage of housing supply in urban areas, further impacting affordability (van Doorn et al., 2019). To address this issue, UN-Habitat has suggested urban densification as a crucial strategy for developing compact and walkable cities (Bibri et al., 2020; Lehmann, 2016; Simon, 2016), in contrast to the modernist, car-oriented planning approach that leads to urban sprawl, which is associated with competing land demands and land loss (Amer et al., 2017). Moreover, the UK government and local authorities have proposed improving land efficiency by increasing residential density around city center transport hubs and suburban areas (Dempsey et al., 2012).

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These urban regeneration and densification projects can effectively curb ongoing urban sprawl, fostering local prosperity and ensuring the long-term sustainability of the urban landscape (Dembski et al., 2020; Mohajeri et al., 2015, 2023).

Increasing housing density in existing built areas and redeveloping non-built land (e.g., urban green spaces, gardens, backyards) are viewed as two common approaches for optimizing urban densification (Abdrabo et al., 2021; Amer et al., 2017; Pauleit et al., 2020). Previous research indicates that housing densification can be achieved through several mechanisms, including: a) increasing density horizontally by constructing new dwellings in backyard gardens (Marique & Reiter, 2014); b) infilling vacant parcels to reduce gaps between urban buildings (Marique & Reiter, 2014); c) demolishing existing low-density buildings and replacing them with higher density high-rise buildings (Amer et al., 2017); d) retrofitting and refurbishing saddle-shaped roofs on top of buildings into more spacious and livable spaces (Pitarch et al., 2020); e) vertical densification achieved by densifying roof stacks and adding floors to existing buildings to accommodate more residents (Amer et al., 2017; Attia, 2015), which optimizes land use and enhances urban space efficiency.

However, existing studies on densification have primarily focused on unidirectional stacking or extensions in either the horizontal or vertical direction (Amer et al., 2017; Deng et al., 2022; Hwang, 2006). Furthermore, many of these approaches (particularly c and d) require demolishing existing buildings before extensions, resulting in significant waste and high construction costs. Typically, solutions involve extending or expanding structures in a single, unidirectional manner rather than a multidirectional approach. The potential for expanding building topology in both horizontal and vertical directions has yet to be thoroughly investigated.

Integrating empirically informed urban and architectural design approaches to guide densification efforts often proves to be a labor-intensive and subjective process at the design stage. Therefore, automated design programs could streamline this process by generating more efficient densification schemes (Eggimann et al., 2021). Recently, some researchers have applied machine learning (ML) to generate architectural and urban designs (Carta, 2022; Gan et al., 2024). For instance, studies have examined the evolutionary dynamics of buildings and analyzed patterns to predict future development in buildings and communities (Hecht et al., 2013; Hussain & Chen, 2018; Mohajeri et al., 2023; Reades et al., 2019). Furthermore, ML is effective for the graphical characterization of building plan generation (Newton, 2019), identifying spatial differences between communities at a regional scale, and classifying their characteristics (e.g.,

housing density) (Mohajeri et al., 2023). Thus, integrating ML models into the densification scheme presents new opportunities to contribute to sustainable housing development by providing valuable data that can inform feedback-based housing typologies and planning methods, enhancing the data-driven decision-making process.

To address the identified research gaps, this study proposes the adoption of generative ML algorithms, particularly Generative Adversarial Networks (GAN), to automatically generate house extension options by extending the roof of a residential building in both horizontal and vertical directions, based on the roof ridge line. The ML-driven house extension framework transitions from voxel-based generation to customizable prefabricated components, enabling rapid and scalable development. This allows house extension solutions to be tailored to different residential building types and neighborhoods, providing efficient residential densification strategies to address urban housing challenges. Using a neighborhood in Walthamstow, East London, UK, as a case study, we demonstrate the significant potential for residential densification in urban areas. This ML-based house expansion method offers a novel approach that can reduce the environmental impact of urban sprawl, contributing to a climate-friendly future while alleviating the housing crisis.

2 Literature review

2.1 Urban sprawl and urban densification

Rapid urbanization has occurred in some regions due to growing populations and economies, leading to the expansion of low-density residential areas, often referred to as urban sprawl (Broitman & Koomen, 2015; Wei & Ewing, 2018). However, studies have shown that urban sprawl can result in spatial inequality, hinder upward mobility, create environmental justice issues, lead to a jobs-housing imbalance, promote social segregation, and exacerbate inner-city problems (Wei & Ewing, 2018; Zhao, 2010). To address these challenges, the New Urban Agenda advocates for preventing urban sprawl through a combination of sustainable land use management frameworks and practices, such as infill development and other expansion strategies (Watson, 2016). This underscores the importance of prioritizing urban densification to achieve appropriate residential densities (Abdrabo et al., 2021; Owusu, 2013). Increasing urban density is a crucial element of UN-Habitat's global urban development strategy, as it can enhance the competitiveness of cities and promote urban economic growth (UN-Habitat, 2013).

Moreover, appropriate residential density can significantly enhance the sustainability of urban and regional prosperity (Dempsey & Jenks, 2010). Consequently, policymakers and researchers increasingly focus on urbanization and its associated factors, such as residential density

(Enserink & Koppenjan, 2007; Kim, 2011). Previous literature has explored the conditions necessary for urban sprawl to occur, as well as the development of various spatial scales and the simulation of non-urban areas that could potentially be transformed into residential areas for densification (Allen & Lu, 2003; Y.-C. Wang et al., 2018).

Nevertheless, the commonly referenced “compact city paradox” illustrates the contradictory relationship between high quality of life, sustainability, and urban density (Neuman, 2005). Urban densification influences multiple qualities that cities value, including aesthetic, functional, economic, operational, environmental, energy, and social aspects. However, the potentially contradictory effects on these qualities must be carefully considered (Cacciaguerra & Gatti, 2015). Achieving a balance between economic, social, and environmental factors is essential for sustainable densification.

2.2 Density and densification

Residential density serves as a critical metric for describing the distribution of residential areas in both horizontal and vertical dimensions. It encompasses how residential areas expand across space and how efficiently they are utilized (i.e., utilization rate) (De Keersmaecker et al., 2003; Mondal et al., 2015). Residential density has two essential aspects: first, the transition from low to high density visually indicates the process of urban densification; second, it is closely linked to various urban characteristics (Broitman & Koomen, 2015; L. Wang et al., 2019). A highly livable city will typically have a reasonable residential density (Kotulla et al., 2019; Mindali et al., 2004; L. Wang et al., 2019). However, numerous factors influence changes in residential density, including geography, environment, natural resources, social processes, and urban planning and development policies (Panagopoulos et al., 2016; Yang et al., 2016).

Case studies on densification have often focused on specific urban areas for design purposes to provide viable options for appropriate density (Broitman & Koomen, 2015). For instance, some studies have assessed the potential for roof stacking at the building level in Brussels, generated 3D urban densification scenarios based on building codes and guidelines, and explored the role of densification in mitigating the urban heat island effect (Amer et al., 2017; Kameni Nematchoua et al., 2020). However, several limitations in existing studies have been identified: a) a focus on the potential for densification from a building perspective; b) limitations in the spatial forms of densification assessment due to reliance solely on high-resolution models from open data platforms; and c) neglect of the assessment of sustainable densification potential (Eggimann et al., 2021). Therefore, it is beneficial to develop alternative methods that can

provide sustainable urban densification strategies at the neighborhood scale in a scalable manner.

2.3 Application of ML in urban studies

Machine Learning has increasingly been utilized in urban analytics, leveraging enhanced computing power and emerging data to improve land use and transportation management. Broadly speaking, ML can be primarily categorized into supervised and unsupervised ML algorithms. Supervised ML algorithms have been trained on various types of datasets that encompass land use patterns, traffic flow, and urban characteristics, facilitating the understanding of relationships between the built environment, socioeconomic factors, and human behavior (Song et al., 2024a, 2024b; Wu et al., 2022). For instance, researchers have employed ML to analyze the evolution of buildings over time and to forecast the location of new developments and the growth of urban communities (Naik et al., 2017; M. Sun et al., 2022). Additionally, ML has been applied to examine complex urban processes such as gentrification, which involves property investment and the role of new residents as developers (Mohajeri et al., 2023). Scholars have also proposed using ML to investigate spatial disparities among communities, categorizing their unique features to inform sustainable housing development plans (Schirmer et al., 2015). In contrast, scholars have mainly used unsupervised ML algorithms, such as various types of clustering algorithms, to identify urban built-form typologies by interpreting urban spatial structure (J. Li & Li, 2024), or through street networks (Yao et al., 2025) or different communities defined by multi-dimensional perceptual qualities (Song et al., 2024a, 2024b).

The application of ML in urban studies encompasses predictive modeling of future trends and optimized resource allocation, employing regression, classification, or time series analysis (Baduge et al., 2022; Jamarani et al., 2024). Furthermore, ML serves as a tool for addressing complex problems involving multiple stakeholders, modeling preferences and constraints through multi-agent reinforcement learning or game theory to facilitate negotiation and solution generation (Liang et al., 2025; Zeleznikow, 2021). This includes generating diverse agent models within urban simulations, utilizing deep learning (DL) algorithms like generative adversarial networks (GANs) or agent-based modeling to create realistic representations of urban actors (Albert et al., 2018; Gan et al., 2024).

2.4 Urban densification and GANs

Research has also increasingly focused on urban densification from both city- and community-scale perspectives using ML for analysis (Amer et al., 2017; Amer &

Attia, 2019; Bibby et al., 2018; Broitman & Koomen, 2015; Mohajeri et al., 2015; Vuckovic et al., 2017; L. Wang et al., 2019). For example, Amer et al. (2017) developed a method to assess building density by stacking roofs, while Bolton (2021) developed a spatial ratio to map the potential for housing density in London, which defines the ratio between existing density and permitted density in each area, usually the maximum allowable density under a specific policy or scenario. Mohajeri et al. (2023) used Random Forests (RF) to quantify the potential of urban densification in the Oxford-Cambridge area, to develop a decision support tool that classifies residential neighborhoods, proposes density scenarios, and estimates housing capacity while also considering the natural capital value of the land to inform more sustainable densification strategies. Despite these advancements, existing research on urban densification is confined mainly to analysis and has not yet fully explored the feasibility of adopting an ML-based generative design framework. Such a framework could be an effective tool for planning and designing sustainable urban densification at the neighborhood scale.

Recently, there has been an expanding body of research on using Generative Adversarial Networks (GAN) (Goodfellow et al., 2014) for land use pattern generation road network design, and neighborhood-scale land use predictions (Fang et al., 2022; Zheng & Yuan, 2021), as well as automated building layout generation for architectural and urban design research (Jiang et al., 2023). For instance, Jiang et al. (2023) developed a site-embedded GAN model to generate building layouts while considering a project's specific urban conditions and design requirements. Zhu et al. (2025) explored using satellite imagery and GANs to predict dockless bike-sharing demand, addressing data limitations in rapidly developing cities. Fang et al. (2022) presented a topography-aware approach for generating road networks that integrate seamlessly with the terrain, while Zheng & Yuan (2021) introduced a generative design method based on neural networks. Complementing these advancements, Gan et al. (2024) developed a UDGAN model, combining improved pix2pix architecture with multi-objective optimization algorithms to generate autonomous urban design solutions.

Given such a missed opportunity, our study advances the state of the art in urban form generation by employing machine learning GAN algorithms to create a novel framework for multidirectional house extensions that integrate horizontal and vertical designs by integrating contextual data. It incorporates specific urban conditions and design requirements into the GAN framework, enhancing the relevance of generated outputs. We develop site-embedded GAN models to produce more

realistic and functional building layouts reflecting urban dynamics. Furthermore, using multi-objective optimization algorithms alongside GANs allows for generating diverse design solutions that balance various urban development goals. These contributions enhance the practical applicability of GANs in urban design, particularly in a novel application in urban densification.

Notably, although ML provides powerful tools for analyzing urban data and modeling complex systems, it is crucial to highlight that humans must remain central to the decision-making process. Ethical oversight is needed to ensure fairness and transparency, with final decisions resting with human actors (Son et al., 2023). The aim is to achieve a balance where AI enhances human capabilities rather than replacing them entirely, particularly in urban research (Son et al., 2023).

3 Case study site

In this research, we selected four low-density neighborhoods in Walthamstow, East London, UK, specifically Valognes Ave, Cogan Ave, Cooper Avenue, Fairview Cl, McEntee Ave, and Millfield Ave, as the preliminary scope of analysis. These areas are characterized by the prevalence of traditional hip-roof housing designs, which constitute a significant portion of the architectural landscape. Notably, their location within the larger London metropolitan area is particularly relevant due to the heightened demand for housing, resulting in a considerable number of building extensions and additions. Specifically, 240 of the 300 households in the four blocks mentioned above have extensions, amounting to 80% of the total households. All the houses in the chart have hip roofs, and detailed extension types for each neighborhood can be found in [Appendix Table 1](#).

As a proof-of-concept study, this research selected Neighborhood 1 ([Appendix Table 1](#)) as a case study site, defined by three streets: Valognes Ave, Cogan Ave, and Cooper Avenue. [Appendix Table 2](#) shows the locations of the extension types for each house, along with detailed photographs. In summary, there are 50 households, of which 42 have had extensions, including 14 rear extensions, eight side extensions, nine corner extensions, and two roof extensions. Additionally, there are 36 independent structures, including enclosures and containers. The following research process grouped the above extension types in a 2D plan and applied them to the training of extension roofs.

4 Data & methods

Our comprehensive framework for generating building extensions and additions comprises five key steps, as illustrated in Fig. 1. 1) Data Collection and

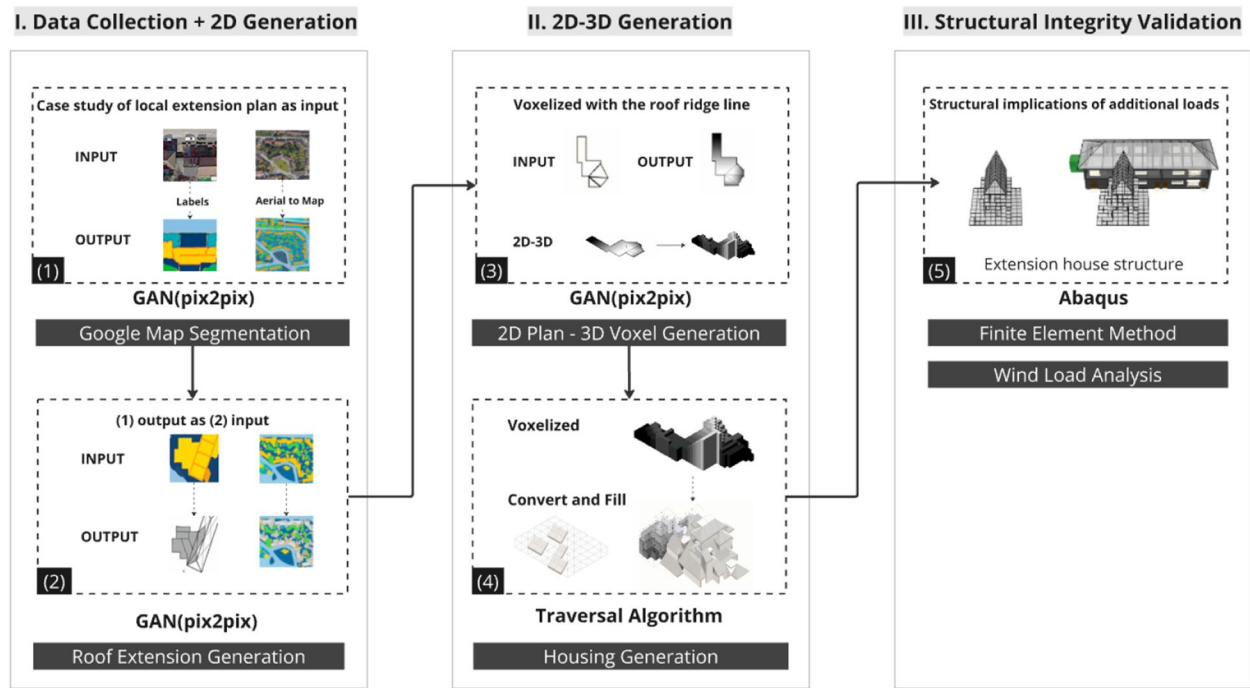


Fig. 1 Research workflow

Segmentation: Google Maps was used to analyze and collect the data for the area with building extensions and additions. The land surface can be segmented using ML, and pix2pix was used to extract the buildings, backyards, gardens, and roads in 2D to generate different color masks and differentiate locations that can be used for building expansion; 2) Roof Plan Generation: Using pix2pix, the ridge Lines of the hip roof, a roof type based on the local architecture, were extended in different directions to generate a roof plan based on the original house design; 3) 2D-to-3D Conversion In pix2pix, a grey-scale map of the roof plan was generated, centered on the ridge Line. The grey-scale map was then voxelized to generate a height value based on its color range. The 2D–3D conversion of the contrast gradient produced a voxel-based building wheel; 4) Structural Generation: The voxel blocks were then transformed into walls in voxel units. Notably, various structures, including roofs, were generated using traversal algorithms; 5) Structural Validation: This study utilized the C3D8 solid element in Abaqus finite element and wind load to conduct numerical simulations. It investigated the potential structural implications of additional loads on building components in the original house structure, verifying the structural validity of the generated extension components.

In this study, the pix2pix model was trained three times in total, employing a hierarchical method to divide the

different data types. This approach allowed for interruptions from unplanned and unexpected changes while waiting for subsequent instructions in the related hierarchy, without interfering with the regular operation of other modules. The front-end and back-end levels of these various tiers were grouped in that sequence. To achieve this, pix2pix, a type of GAN model, was determined to be the most suitable option because its front-end map recognition mechanism directly captures the user's changing needs (Benjdira et al., 2020).

4.1 Map segmentation

We analyzed land use, roof types, and extension types for houses on 14 streets (four blocks) using Google Maps. We employed a pix2pix model, a conditional GAN for image-to-image conversion, to transform the Google satellite maps of Valognes Ave, Cogan Ave, and Cooper Avenue into maps with different colors representing various components. Each map was cropped into 256×256-pixel patches. Using ML, this process involved manually segmenting elements in the maps, such as houses, roads, backyards, front yards, trees, public spaces, and undeveloped land. This simplification retained the elements that directly affected future planning while reducing the complexity of the geographic content. The pix2pix generation process, which involved a literal translation with one-to-one correspondence, gradually improved as training epochs and datasets increased, leading to more accurate

learning and adaptive capabilities (Isola et al., 2016). In the model training process, the learning rate (lr) was set to a default value of 0.0002, employing the Adam optimizer for parameter optimization. The first momentum coefficient (beta1) of the Adam optimizer was set to a default value of 0.5. For input images with dimensions of 256×256 pixels, considering GPU memory constraints and model complexity, this study adopted a micro-batch training strategy with batch size = 1. While this setting increased gradient variance during training, it effectively controlled memory usage. The discriminator and generator of the Gans were trained simultaneously, and 50 training epochs were performed in this section, the model achieved stable convergence with satisfactory accuracy.

4.2 Roof extension generation

Next, the model was used to investigate how the local extension types were connected to the original houses, and the study of these extension types was used as the target output for the next-step ML training. Notably, the input dataset for the pix2pix model consisted of the map from the outputs in 3.3, and the output was a two-dimensional map of the roof ridge that was obtained by expanding the original roof ridge line of the local hip roof and subsequently generating the building footprint with a new roof ridge line. The pix2pix model was trained on a dataset of 300 images (256×256 pixels), which were augmented through random flipping and rotation to improve generalization. We found increasing the number of training epochs to 600 led to better convergence, as was monitored by the stabilization of generator and discriminator losses. This extended training duration compensated for the limited initial data volume while avoiding overfitting through early stopping that was based on validation set performance.

4.3 Building generation by 2D Plan to 3D voxel

Subsequently, the roof plan was voxelized with the roof ridge line to generate a grey-scale roof map. The pix2pix model was further trained using the output from 3.2 as the input. The output was a grey-scale map, with a white color centered on the roof ridge line and a gradient of grey transitioning to the black edges. After obtaining the grey-scale roof image, it was converted to a voxelized plane, and the height was generated according to the color scale range. The lighter the color in the plan, the higher the height of the space represented. The voxelized grey-scale image of the roof was transformed into a 3D voxel stack with decreasing heights. Finally, a traversal algorithm transformed the wall used for the 3D voxel into a voxel base. Herein, we algorithmically listed all possible combinations and automatically filtered out the optimal solution for filling the 3D voxel with various components.

4.4 Structural analysis

Finally, to validate the structural feasibility of this workflow, the study employed the C3D8 solid element in Abaqus finite element software to conduct simulations to explore the potential impact of additional loads on existing building components for the original house. It can comprehensively consider factors such as load transfer paths and structural deformation coordination and accurately predict the impact of increased loads from newly constructed components on the original house structure's bearing capacity, deformation, and stress distribution (Illampas et al., 2014). The analysis ensures that the stresses in the structure will not exceed the yield strength of the materials under normal conditions, considering all structural elements as ideal elastic materials. The main structural parameters were as follows: for the existing building structure: elastic modulus of 34.5 GPa, Poisson's ratio of 0.2, and 2500 kg/m^3 density (i.e., typical masonry, concrete, or a combination). For the extended structure, the settings were: elastic modulus of 10 GPa, Poisson's ratio of 0.2, and 500 kg/m^3 density.

4.4.1 Finite Element Method (FEM)

The contact conditions were defined as tied contact between the roof and the main structure, ensuring no significant slip under normal conditions and satisfying the accuracy requirements. The interface between the new and existing building structures was modeled using a combination of contact conditions to simulate the interaction at the interface. In the normal direction, a hard contact formulation was employed to capture the non-penetration behavior at the interface. In the tangential direction, a penalty function represents the frictional behavior, with a friction coefficient set to 0.5. This contact modeling approach allows for the realistic simulation of the interface conditions between the new and existing structures, ensuring the appropriate transfer of stresses and forces across the interface. The entire model was loaded with gravity loads, and following the GB50009-2012 building code, a uniformly distributed live load of 2.0 kN/m^2 was applied to the floors. By analyzing the structural response under different load conditions, potential safety hazards can be promptly identified, supporting the necessary reinforcement measures for the generated building extension options.

4.4.2 Wind load analysis

Lastly, this research further takes into account the dynamic load, particularly the wind forces in case of wind uplifting risks. According to BS EN 1991-1-4 of the European Committee for Standardization, the basic wind speed of 21 m/s for wind-resistant design

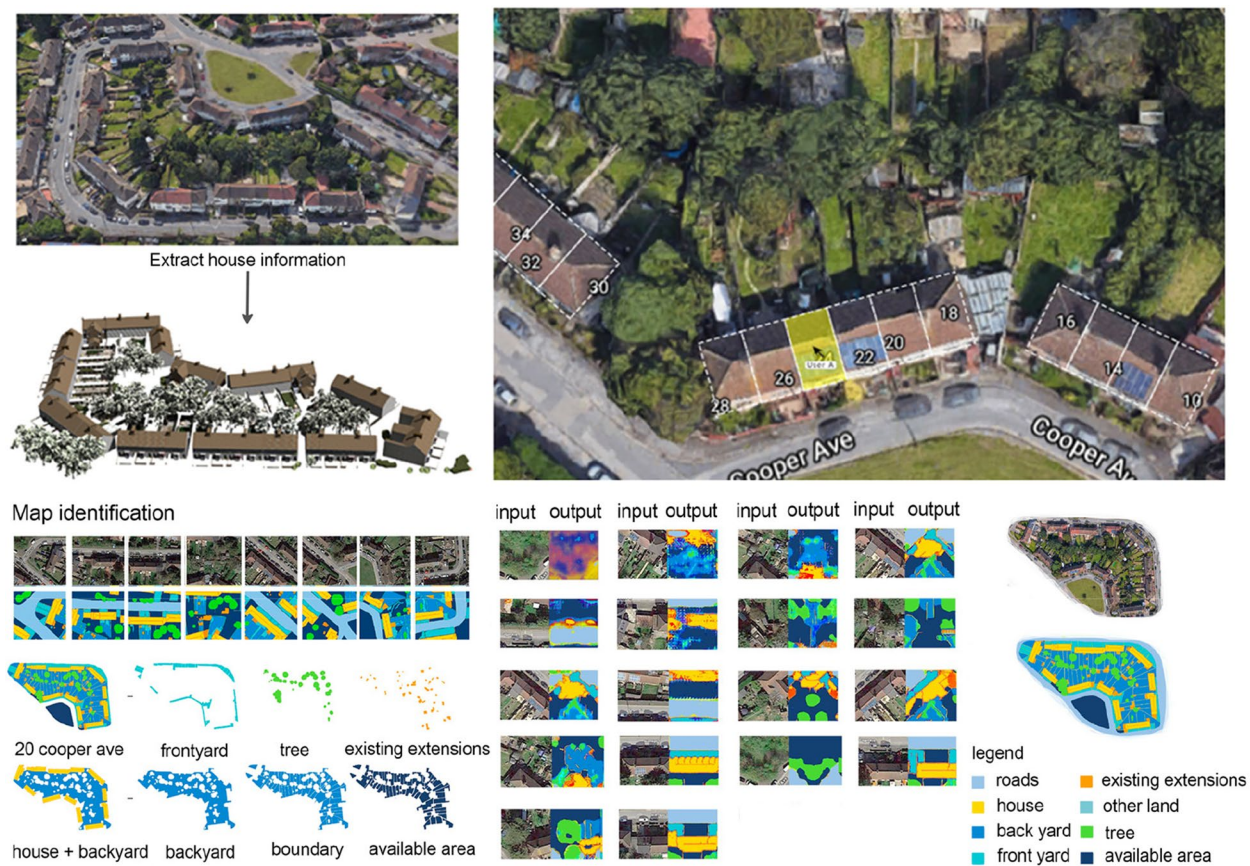


Fig. 2 Google map segmentation

in London, which includes a safety factor (European Committee for Standardization (CEN), 2005). Urban or suburban areas with numerous obstacles such as buildings, trees, and other structures as obstacles terrain category IV, exposure coefficient 1.2, terrain coefficient 1.0, design wind pressure ≈ 380 Pa. In building design and wind load calculations, terrain factors are used to ensure that structures can withstand the wind forces in a specific area, ensuring safety and durability (European Committee for Standardization (CEN), 2005). The predominant wind direction in London throughout the year is from the southwest (Lapworth & McGregor, 2008), making the northeast facade of the building the windward side. Load tests are conducted on the eastern and northern walls in Abaqus.

5 Results

5.1 Google map segmentation

For map segmentation, a pix2pix model was meticulously constructed using Google Maps images of Valognes Ave, Cogan Ave, and Cooper Avenue as the input to the model; we extracted house information and corresponding manually labeled color-coded map images of various

features as the output (Fig. 2). Appendix Fig. 9 provides the list of segmented features trained in the pix2pix algorithm. The color-coded maps (Fig. 2) were used to train the recognition of different spatial feature patterns in different areas (e.g., residential buildings, gardens, backyards, and open spaces).

5.2 Extension generation by the roof ridge line

After obtaining a potential area representing the footprint of the house extension, the following model transformed the potential backyard area into a roof footprint. The target output dataset was manually labeled and collected from a study of extensions in the region, with the default selection being the hip roof, which is widely used in residential housing in the UK. In Appendix Table 2, the inventory of extension types reveals that the extensions are primarily concentrated at the rear of the houses, in the backyard, and on the roof. These extension types were integrated into a database as the target for the next training session (see Fig. 3) in pix2pix.

To achieve this, we utilized the segmented maps as input for our trained Pix2Pix model, which was adapted to generate 2D roof maps expanded along the original

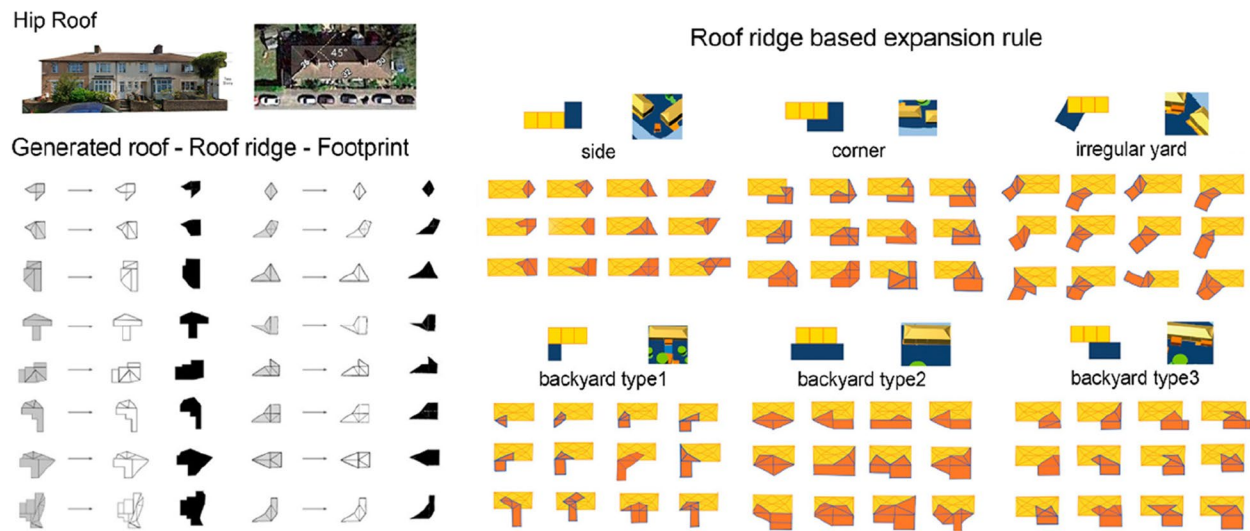


Fig. 3 Case study of local extension as input

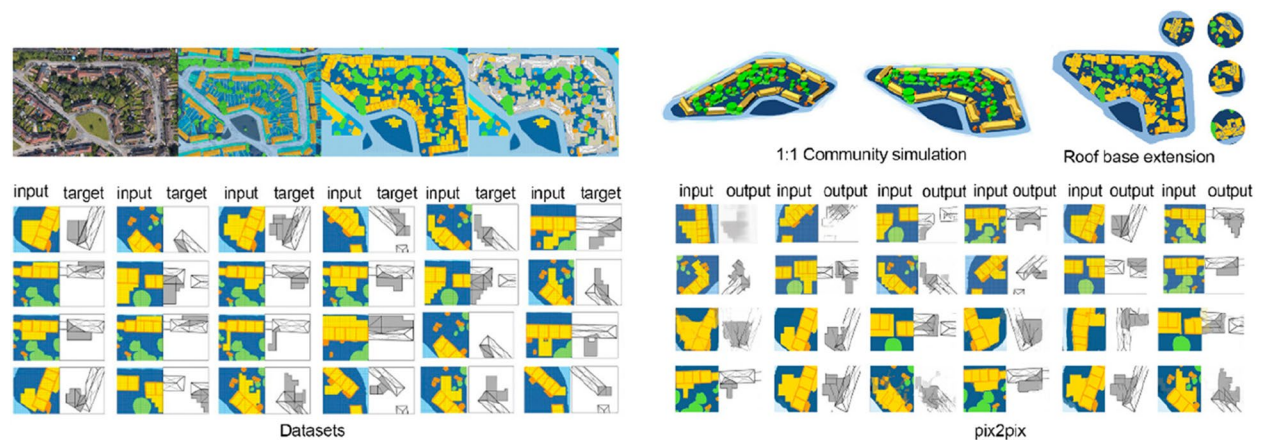


Fig. 4 Generated extension of roof ridges

ridge Line based on learned house extension types. We manually created a diverse test dataset consisting of 300 images, representing various roofing styles and conditions. Data augmentation techniques such as random cropping and flipping were applied to enhance the training dataset, helping to reduce overfitting and improve the model's generalization. The Pix2Pix model was trained for 600 epochs, a number determined through preliminary experiments showing effective convergence of loss values. This training duration was essential for the model to capture the complex features and variations present in the roofing structures, as illustrated in Fig. 4.

Notably, the simulated generation may be biased in its representation because the target output dataset was formed from a study of extensions in the area using the hip roof by default. The initial study of the extension geometry prototype provided a particular 3D model of

the local house, which may have Limited the diversity of generation outcomes. Fortunately, some space remains between conversions when compressing the 3D information into 2D shapes. Machine-generated ridgelines often exceed the original pre-determined values. The potential 3D form was abandoned in the later integration of the dataset to reduce the influence of the researcher's perspective on the results. Instead, only the relationship between line segments and planes was considered. This machine-oriented thinking suggests a changing role of the researcher in the generative design planning process.

5.3 2D pixel to 3D voxel base building generation

Figure 5 illustrates how these 2D roofs were converted into 3D voxels and how they relate to the original house. The roof plan with the ridge Lines was used as the input

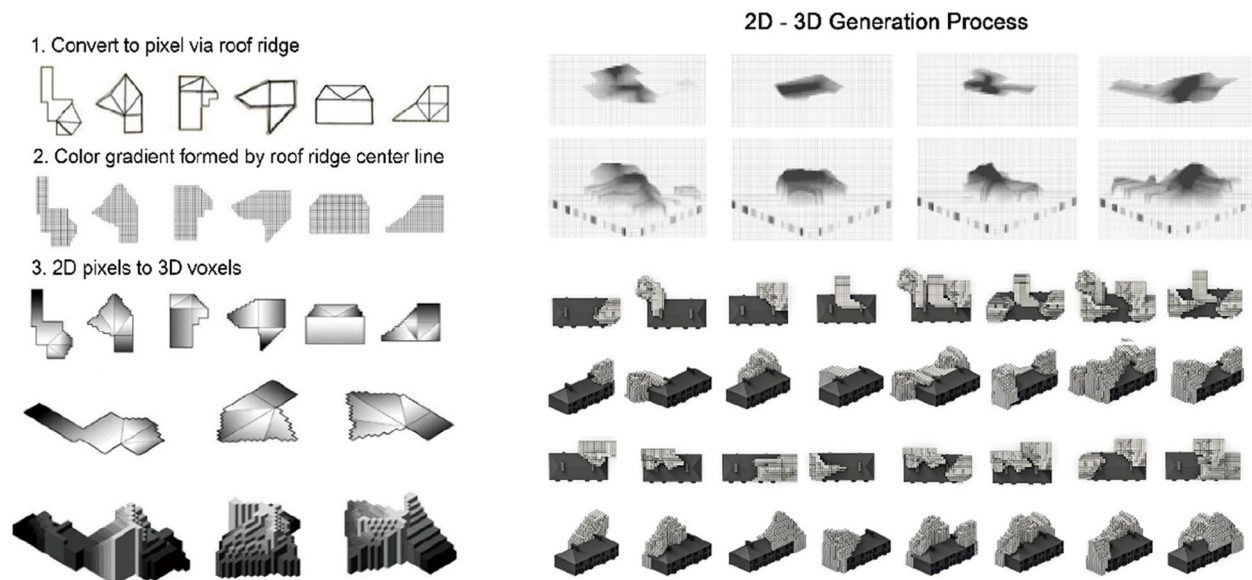


Fig. 5 2D–3D voxel stack generation

to the model, with the grey-scale roof map as the output. The plan and height maps were converted from 2D pixels to 3D voxels, generating voxel stacks of different heights based on color scale increments.

5.4 3D Generation rules and algorithms

After obtaining the 3D voxels from the roof image plan, the voxels were replaced with walls to generate the building structure. Step 1: The coordinates were used to determine the number of voxels occupied by the portion of the roof shown in the image (see Fig. 6). Step 2: The corresponding substitution relationship between the roof and the voxels was established. When generating the roof sections, different classes of roof parts were extracted based on the presence of neighboring voxels. Step 3: The roof elevations were randomly generated under restricted conditions. We employed a traversal algorithm to determine the direction with the most voxels at the current voxel position. The voxels were then traversed in a bottom-up, right-to-left direction, continuously cycling through each layer until all voxels were filled. Figure 7a presents three examples, showing the process diagrams generated for each example. Figure 7b displays the original house structure alongside the newly generated structure.

5.5 Generated results for building extension

When applying the workflow to the entire case site, Fig. 8 presents the generated house extensions integrated with the actual street in the selected blocks, visualizing the outcome of this 3D generation process in the blocks of Valognes Ave, Cogan Ave, and Cooper Avenue.

5.6 Structural analysis

5.6.1 Finite Element Method (FEM) static analysis

The study further reported two experiments to investigate the structural implications of building extensions on existing structures for demonstration purposes. In the first experiment, the extension structure was in the rear yard, i.e., a rear extension. The interaction between the roof components should have been addressed, and the analysis focused on the binding behavior between the main structure and the extension. The detailed results are presented in Appendix Fig. 10. For the Rear Extensions scenario (Experiment 1), the maximum stress value in the structural elements is 36.55 MPa, and the maximum displacement is 0.1076 mm. The yield strength of the clay bricks is 20–35 MPa, mortar is 2–7 MPa, and Hardwood Lumber is 15–30 MPa. Results indicate that the maximum stresses in the structural elements are within the yield strength ranges of the clay bricks and hardwood lumber but exceed the yield strength of the mortar. The small maximum displacement suggests good overall structural performance and serviceability. It indicates that the structural integrity of the integrated system, comprising the main building and the rear extension, is maintained, and the design satisfies the serviceability requirements with a large safety margin. The second experiment considered an extension structure in the side-rear area, i.e., a combination of rear and side extensions. In this case, the analysis included the interactions between the roof components, and the overall model was subjected to gravity loads (see Appendix Fig. 11). The maximum stress levels observed in the overall structural model, including the roof components

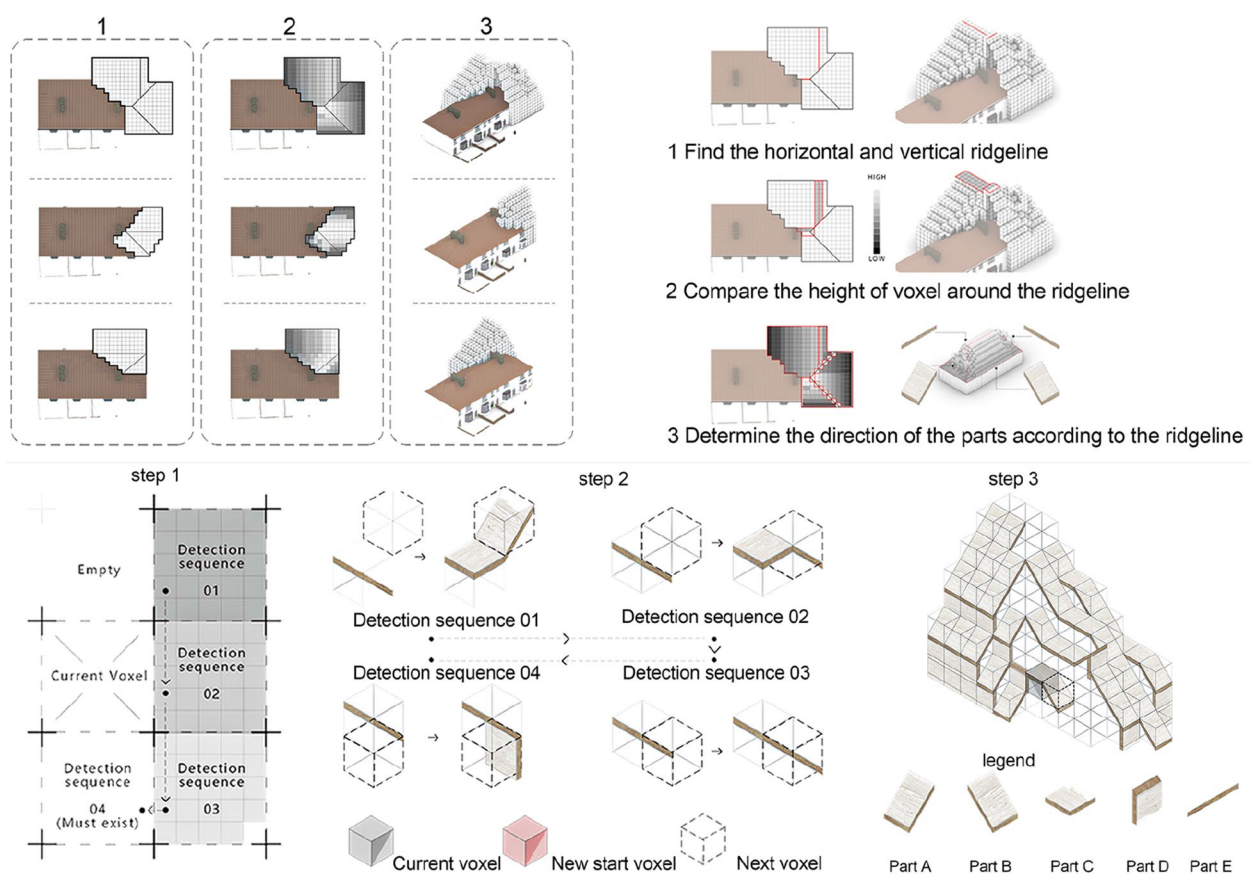
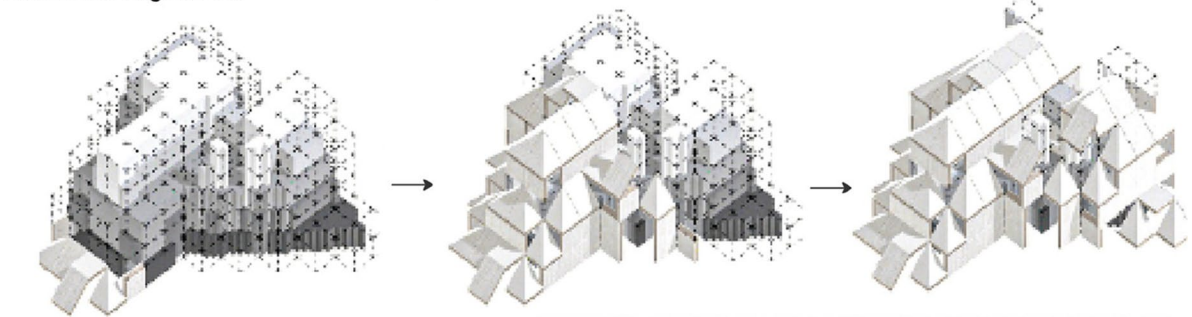


Fig. 6 Corresponding substitution rules of voxel replacement

a. Traversal Algorithm



b. Voxel Base House Generation

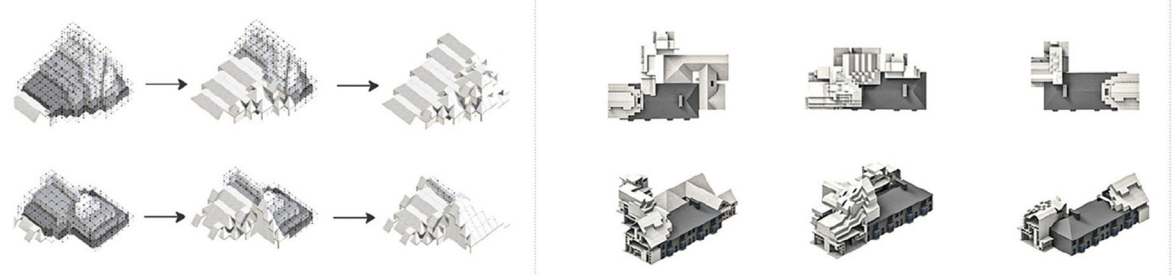


Fig. 7 a Voxel-based building generation by traversal algorithm, (b) The generated new structure and the original house



Fig. 8 Design Proposal (a) Actual streets by Google map (b) Generated results

and the combined extension, is 7.917 MPa, and the maximum displacement is 5.456 mm. This stress magnitude is considerably lower than the material's yield strength of clay bricks: 20–35 MPa, mortar: 2–7 MPa, and hardwood Lumber: 15–30 MPa; the maximum displacement of 5.456 mm is relatively small, suggesting good overall structural performance and serviceability and demonstrating that the integrated structure can safely withstand the applied gravity loads without exceeding the allowable limits.

These two experiments show that the structural solutions adopted for the building extensions, whether located at the rear or a combination of rear and side, can effectively transfer the loads and maintain the structural integrity of the modified building system. The analysis has confirmed the feasibility of our ML-based generative design approach, with the selected structural elements being appropriate to meet the necessary safety and serviceability requirements.

5.6.2 Wind load analysis

An assessment of the wind resistance performance of two houses was conducted. For House 1, the maximum stress recorded was 1.172 MPa, with a maximum displacement of 4.3 mm and a maximum strain of 0.00003702. In contrast, House 2 exhibited a maximum stress of 1.493 MPa, a maximum displacement of 1.846 mm, and a maximum strain of 0.00004335.

The maximum stress values for both houses fall below typical yield strength thresholds for standard construction materials for masonry, indicating a safe design not less than 13.8 MPa (Brick Industry Association, 1992).

The recorded displacements of 4.344 mm and 1.846 mm are generally considered acceptable within most architectural design standards, particularly concerning wind resistance. Additionally, the strain values suggest that neither structure will experience excessive deformation under wind loads. Overall, both houses demonstrate satisfactory wind resistance performance, with maximum stress, displacement, and strain values remaining within conventional design standards. It is advisable to confirm that these values are within the safety limits of the selected materials and to consider applicable building codes and standards to ensure comprehensive compliance with wind resistance design requirements (see Appendix Fig. 12).

6 Discussion

6.1 Feasibility of ML-based generation workflow

Existing studies on residential densification have primarily focused on a unidirectional approach, either increasing building density horizontally or vertically (Amer et al., 2017; Attia, 2015; Marique & Reiter, 2014; Pitarch et al., 2020). However, the potential for synergistic expansion of building topology in both horizontal and vertical directions remains scarce. This study addresses this gap by developing a novel approach that leverages GAN to generate building extensions in both dimensions, which offers an efficient and less subjective alternative to the current labor-intensive manual design process by integrating housing typologies within existing neighborhoods.

Results from the study in Walthamstow, East London, demonstrate the potential of this method for urban densification and optimizing usable land within urban boundaries, thereby mitigating urban sprawl and alleviating the housing crisis. Notably, when trained adequately with labeled inputs, GAN models like pix2pix can automatically identify and classify geographic information and architectural features in neighborhoods, simplifying complex spatial data to provide more intuitive inputs for planning and design. The automatic generation of 2D floor plans and 3D models for different expansion scenarios significantly improves planning and design efficiency. Automated data processing and design optimization offer homeowners expansion alternatives that meet their specific needs, enhancing their quality of life. This data-driven workflow also supports community planning to address residents' shifting housing demands and improve overall community adaptability (P. Sun et al., 2023).

While previous research has explored the use of machine learning in architectural design, particularly for tasks including building layout generation and energy performance optimization, this study's innovation lies in its holistic approach to urban densification by simultaneously considering horizontal and vertical expansion possibilities within existing urban contexts. This complements the existing literature that typically addresses these dimensions in isolation. Furthermore, our approach uniquely integrates a GAN-based model to generate design options that are spatially efficient and sensitive to the existing architectural style and urban fabric, promoting seamless integration and community acceptance. The novelty of our work also stems from the development of a practical workflow that urban planners and homeowners can readily adopt. By automating the generation of design alternatives and providing clear visualizations, we empower stakeholders to make informed decisions about building extensions, facilitating more sustainable and equitable urban development.

6.2 Practical considerations

After finalizing the generated option, in the construction phase, modular prefabrication techniques can be applied to enable rapid assembly of temporary housing structures, improving efficiency and quality while reducing the environmental impact of the building (Y. Li, Yu, et al., 2023a, 2023b). It is also essential to clarify the construction techniques and structures that are most suitable for permanent or temporary buildings during the pre-design phase (W.-C. Wang & Lim, 2014). It is worth noting that our method has taken this into account; for instance, it proposes modular components like a roof in the generation process from 2 to 3D voxels, which allows for later optimization potential in the construction stage. By

providing a flexible configuration that enables modular construction, this methodology can significantly enhance the cost-efficiency and accessibility of housing solutions, particularly in affordable housing projects (Khan et al., 2024; Wai et al., 2023). One of the significant innovative aspects of our research lies in integrating ML algorithms to automate the design of these modular components, ensuring that they are structurally sound and contextually relevant to the existing urban fabric. This data-driven approach allows for rapid adaptation to local needs, promoting more sustainable urban densification strategies.

The automated generation of building layouts and components can streamline construction, reducing repetitive labor and empowering AI-assisted design practices. Specifically, prefabricated modular technology empowers efficient and high-quality housing expansions, which reduces construction disruptions to existing buildings and environments and lowers the expansion process's energy use and environmental impact (Casini, 2021; Y. Li, Hu, et al., 2023). The modular nature of the extensions, combined with the automated design process, should allow for faster and more scalable deployment of these house extension solutions. This is crucial for promptly impacting urban densification and the housing supply (Oorschot & Asselbergs, 2021). Our approach addresses existing gaps in the literature by demonstrating how ML can enhance the precision and customization of modular designs, thus supporting urban planners in creating more resilient communities. This contribution is particularly significant as it moves beyond traditional methodologies that often overlook the dynamic nature of urban environments.

Moreover, finite element analysis simulates the additional loading impacts of new construction on existing structures, allowing the timely identification of potential safety risks to evaluate the generated options structurally (Haldar & Mahadevan, 2000). The static analysis of the structural elements in both experiments provides a thorough understanding of the structural implications and challenges associated with integrating building extensions, particularly in load transfer, stress distributions, and the overall integrity of the modified structural system. Integrating such analysis into the overall workflow can help inform design guidelines and decision-making processes for homeowners and local authorities when considering building modifications and extensions, ensuring the integrated system's structural integrity and balancing long-term performance.

6.3 Implications

6.3.1 Implications on planning and design workflow

It is worth noting that all training data in this study were garnered from existing extensions in the studied neighborhood, which mainly adopt the hip roof design. The study uses ML to generate multiple visualized hip-roof

scenarios (as default) for house extensions, which can inform more targeted urban planning policies. This innovative approach represents a significant advancement in the integration of ML within urban and architectural design disciplines, enabling practitioners to visualize and assess the impact of various extension scenarios on the urban landscape in a data-driven manner. At the same time, one may question the applicability of such a method to other roof types. [Appendix Fig. 13](#) lists five common roof types in the UK. Since the generation logic developed in our workflow is based on the transition from voxels to modularization, it can be later adjusted according to different roof types if the module parts are replaced within the voxel block. This flexibility in the framework enhances its applicability across diverse architectural styles, thereby broadening its relevance and utility in urban densification efforts. Therefore, the expansion workflow proposed in this paper can be effectively generalized to other roof types.

Recent studies have applied GAN-based models and machine learning across urban design scales, from site-specific building layouts to autonomous road network generation and generative architectural methods (Fang et al., 2022; Jiang et al., 2023; Zheng & Yuan, 2021). However, their industrial application still faces limitations, particularly regarding scalability and adaptability to site-specific conditions.

Notably, when combined with the design generation methodologies explored in this study, the predictive capabilities of GANs could unlock new opportunities for optimizing the densification of the built environment while minimizing environmental impacts. Building upon these successes, a clear pathway emerges for extending the framework's application to a broader range of real-world projects, from affordable housing developments to urban infrastructure planning and community-driven design initiatives. This innovative integration enhances design efficiency and aligns closely with contemporary sustainability goals by enabling more responsive and adaptive urban planning strategies.

While the current framework may not yet directly address the challenges of building additions and retrofits on existing structures, the foundational work presented in this study lays the groundwork for future advancements in this domain by establishing a robust methodology that can be adapted and refined for various architectural contexts. Theoretically, the ability to generate novel land use patterns and building forms using GAN-based ML methods presents new possibilities for optimizing urban space utilization and achieving more efficient densification strategies. By embracing the opportunities this research offers, stakeholders in the built environment can harness the power of ML and

computational design to deliver more cost-effective, sustainable, and community-centric solutions.

6.3.2 Implications on public engagement

Furthermore, the predictive capabilities of the GAN models can be leveraged to improve the efficiency and responsiveness of various government-led development initiatives. Moreover, beyond the technical applications, the proposed framework also holds the potential to facilitate enhanced public participation in the design and development process. Leveraging the ML-based generation approach, residents are now empowered to collaborate with designers to customize these extension proposals based on their unique needs and explore possibilities for expanding their living space. The research team can create interactive interfaces that allow community members to explore and provide feedback on proposed interventions by training the GAN models to generate diverse design scenarios (Chew et al., 2024; Hughes et al., 2021; Wu et al., 2022). This can simplify the public engagement process, foster greater transparency, and empower citizens to be more active in shaping their built environment.

However, translating these technologies into concrete practices that drive urban development requires overcoming a series of technical, institutional, and social barriers (Bibri, 2021; Green, 2019; Regona et al., 2024). This necessitates collaborative efforts between researchers, designers, and decision-makers; concurrently, we cannot overlook the advancements in applying these methods within architectural and urban design education; these innovative explorations indicate that data-driven and computational design techniques are gradually incorporated into design practice, where a balanced understanding of the limitations of these emerging techniques is coupled with a continuous exploration of pathways to breakthrough to facilitate the realization of sustainable urbanization, laying the foundation for future application developments (Bibri, 2021; He & Chen, 2024; van Ameijde, 2023).

Research shows that integrating ML and large language model technologies can transform public engagement in urban planning by creating more inclusive participatory processes (Xu et al., 2024). These AI tools democratize design by providing user-friendly interfaces for real-time visualization of planning scenarios. This allows human planners to focus on value-based decision-making while AI handles data-intensive tasks (Fu et al., 2024; Xu et al., 2024). However, challenges remain regarding algorithmic transparency, the digital divide, and the need for robust verification mechanisms to ensure the validity of AI-generated proposals (Fu et al., 2024). When implemented thoughtfully, these tools can enhance the quality and quantity of public participation, making planning more

responsive to community needs. Notably, they should support rather than replace human decision-making. If misused and lacking oversight, such AI tools may introduce biases in public feedback, potentially marginalizing voices from underrepresented groups and exacerbating urban inequalities in planning processes (Fu et al., 2024).

6.4 Limitations and future work

However, this study is not without limitations. First, our research mainly centers on developing a feasible automatic generation workflow, while it can benefit from adding one extra step that solicits residents' preferences on generated options. Previous urban studies have utilized visual surveys to reveal housing preferences (Song et al., 2023) and people's visual perceptions of the built environment (Dubey et al., 2016). Understanding the reasons for specific preference patterns offers a more holistic view of urban intensification at the individual level. Furthermore, relying on GAN models may introduce biases in the training data, potentially leading to design solutions that perpetuate existing inequalities or fail to adequately address the needs of diverse communities. Future research should explore techniques for mitigating these biases and ensuring that the ML-generated designs are equitable and inclusive.

Second, the study should have considered the presence or absence of historical buildings before proposing a housing extension. Preserving historic architecture is an essential factor in future urban densification projects. To assess residents' attitudes toward housing expansion in neighborhoods with historical buildings, the government and designers can present various proposals for housing expansions in community meetings and conduct questionnaires (Tsai et al., 2021).

Third, it is uncertain how the proposed building extensions would consider the aesthetic standards of architecture within the specific context. However, Machine learning techniques could be leveraged to enhance the consideration of aesthetic factors in the design process, for instance, in designing architectural facades. Notably, deep learning models trained on historic and culturally significant building databases could mimic the architectural styles and features of existing structures in a neighborhood (Kumar et al., 2020). This could then further inform a culturally sensitive design of new extensions, ensuring the proposed housing expansions not only suffice the densification needs but also maintain or enhance the neighbors' visual coherence and architectural integrity.

Fourth, due to resource and time constraints, we were unable to conduct stakeholder interviews. These interviews could have provided deeper insights into the perspectives and needs of various stakeholders. Additionally, a cost–benefit analysis was not included in this study, which could have assessed the relationship between the resources required for implementation and the potential benefits. We plan to incorporate these

elements into our workflow in future research to provide a more thorough analysis and conclusions.

Lastly, given the resource constraints of this research, the explicit focus on hip roofs may limit the findings' generalizability to other types of roofs and housing structures. Nevertheless, the methods adopted in this research can be viewed as a starting point and a baseline, further involving case studies in different urban contexts. Future studies could explore the adaptability of our approach to a wider range of roof typologies and housing structures across diverse geographical contexts. We also plan to investigate how machine learning models and design algorithms can be further refined and fine-tuned to accommodate the unique characteristics and requirements of various building forms on a larger scale.

7 Conclusion

This research introduces a novel analytic and design workflow for urban densification literature: an automatic generation framework that integrates machine learning techniques, particularly GANs, as a sustainable development approach to address the pressing housing crisis caused by growing urban populations. Utilizing a low-density neighborhood in London as a case study, the framework demonstrates the potential of multi-directional roof extensions as a viable strategy for residential densification. The findings suggest that our proposed method can be seamlessly applied to diverse neighborhoods and adapted to local housing characteristics, showcasing its contextual responsiveness and broad applicability. Additionally, the research underscores the importance of harmonizing design principles with advanced machine learning methods to optimize urban development processes. Grounded in a rigorous structural analysis of housing expansion patterns in East London neighborhoods, the case study validates the reliability of machine learning and GANs in generating automated extensions of residential roofs. This computerized workflow presents a viable approach for city-scale housing expansion, providing authorities and urban design professionals with a novel perspective for efficiently implementing sustainable, data-driven decision-making solutions.

The proposed approach offers a holistic and technologically driven solution for addressing the housing crisis, bridging the gap between traditional urban and architectural design and computational intelligence. By integrating contextual characteristics, data-driven optimization, and scalable implementation, this research paves the way for a more sustainable and context-responsive future for urban housing development. The findings hold promising implications for policymakers, planners, designers, and housing authorities as they navigate the complex landscape of ongoing urban growth and heightened housing demands.

Appendix

Table 1 House Extension in 4 blocks of Walthamstow, East London

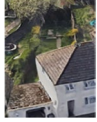
Location	Ranges	Number of houses	Housing Extensions	%	Specific Typological Extensions				
					Rear	Side	Corner	Roof	Building, enclosure, or container
1	 Valognes Ave; Cogan Ave; Cooper Avenue	56	42	75	14	8	9	2	36
2	 Fairview Cl; Higham Hill Rd; B179; Millfield Ave	93	51	55	21	2	1	8	30
3	 McEntee Ave, B179; Lawrence Ave; Garnett way	72	65	90	32	6	1	5	33
4	 Millfield Ave; Lyne Cres; Guildsway; Williams Ave	79	74	94	46	5	8	3	28
TOTAL		300	232	77	113	21	19	18	127

Notes: Neighborhood 1: Valognes Ave, Cogan Ave, Cooper Avenue; Neighborhood 2: Fairview Cl, Higham Hill Rd, B179, Millfield Ave; Neighborhood 3: McEntee Ave, B179, Lawrence Ave; Garnett Way, Neighborhood 4: Millfield Ave, Lyne Cres, Guildsway, Williams Ave

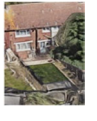
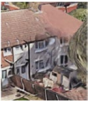
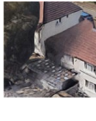
Table 2 Inventories of specific typological extensions in our study area

(a) Valognes Ave							
Study areas	Valognes Ave 1	Valognes Ave 11	Valognes Ave 13	Valognes Ave 17	Valognes Ave 19	Valognes Ave 21	Valognes Ave 23
Latitude and longitude	51.600220235303624, -0.03558404767677032	51.600267787085905, 0.03523412897872555	51.60043048116458, -0.03512585070434429	51.600467273843826, -0.03510671056318924	51.6005748749509, 15.0.03514565876 439779	51.60066434231022, 0.03514302553109465	51.60071101599512, -0.03514521473388256
Snapshots							
Rear extensions					√		
Side extension	√	√	√				
Corner extension	√	√	√		√	√	
Roof extensions							
Building, enclosure, or container			√	√	√		√

(a) Valognes Ave

Study areas	Valognes Ave 25	Valognes Ave 29	Valognes Ave 33	Valognes Ave 35	Valognes Ave 37	Valognes Ave 39
Latitude and longitude	51.600792509344515, 0.0351700295174718	51.600935642602934, -0.03520818260282716	51.601003471785624, -0.03535201416745535	51.601035365106355, -0.03539294331995548	51.60106561320197, -0.03545676257309926	51.601114181270134, -0.03552256826313023
Snapshots						
Rear extensions			√		√	
Side extension						√
Corner extension		√				
Roof extensions						
Building, enclosure, or container	√	√	√	√	√	√

(b) Cogan Ave

Study areas	Cogan Ave 2	Cogan Ave 4	Cogan Ave 6	Cogan Ave 10	Cogan Ave 12
Latitude and longitude	51.60114112951181, -0.036920900957585906	51.60111844946429, -0.036823754563730646	51.601134808130254, -0.036751920604252745	51.60109246902854, -0.03652636875502721	51.601100797372546, -0.03645417272618641
Snapshots					
Rear extensions				√	
Side extension	√				
Corner extension					
Roof extensions					
Building, enclosure, or container	√	√	√	√	√

(b) Cogan Ave

Study areas	Cogan Ave 14	Cogan Ave 16	Cogan Ave 20	Cogan Ave 22	Cogan Ave 26
Latitude and longitude	51.6011034931514, -0.03637145287303537	51.60111400936585, -0.03627506545179216	51.601118428512265, -0.03609928263005778	51.601116002064664, -0.03597522658595262	51.60111929553716, -0.03579459179459763
Snapshots					
Rear extensions		✓		✓	✓
Side extension			✓		
Corner extension					
Roof extensions			✓		
Building, enclosure, or container	✓			✓	

(c) Cooper Ave

Study areas	Cooper Ave 6	Cooper Ave 5	Cooper Ave 4	Cooper Ave 8	Cooper Ave 10	Cooper Ave 16	Cooper Ave 18
Latitude and longitude	51.60039071835889, -0.03572066679457797	51.60042778998862, -0.03573147330974597	51.600474204029204, -0.035749161142699874	51.60054479104254, -0.03578433670466819	51.60066922832431, -0.03584720720980041	51.60070075312645, -0.03612430272364615	51.600706335024235, -0.036295412607623236
Snapshots							
Rear extensions	✓	✓			✓	✓	
Side extension							✓
Corner extension					✓	✓	
Roof extensions							
Building, enclosure, or container	✓	✓	✓	✓	✓	✓	✓

(c) Cooper Ave

Study areas	Cooper Ave 20	Cooper Ave 24/26	Cooper Ave 27	Cooper Ave 28	Cooper Ave 30	Cooper Ave 32
Latitude and longitude	51.600692037908345, -0.03637812817635934	51.600687163524455, -0.036512343107767214	51.600652550484696, -0.036579186155287616	51.60065518271567, -0.0366574176913998	51.600746768082274, -0.036909830124182634	51.60075790004776, -0.037022476804572255
Snapshots						

(c) Cooper Ave						
Study areas	Cooper Ave 20	Cooper Ave 24/26	Cooper Ave 27	Cooper Ave 28	Cooper Ave 30	Cooper Ave 32
Rear extensions				√		√
Side extension						
Corner extension				√		
Roof extensions						
Building, enclosure, or container	√	√	√	√	√	√
(c) Cooper Ave						
Study areas	Cooper Ave 34	Cooper Ave 36	Cooper Ave 38/40	Cooper Ave 41	Cooper Ave 42	Cooper Ave 44
Latitude and longitude	51.600813231354934, -0.03705976104853952	51.600855071243366, -0.03711670290838312	51.600996210888674, -0.0374185176137604	51.6010562728974, -0.03736680325030121	51.60109634903957, -0.037346719350231	51.60119210443372, -0.03726497118892926
Snapshots						
Rear extensions					√	
Side extension						√
Corner extension						
Roof extensions						√
Building, enclosure, or container	√	√	√	√	√	√

Note: √ indicates the presence of corresponding characteristics for specific housing

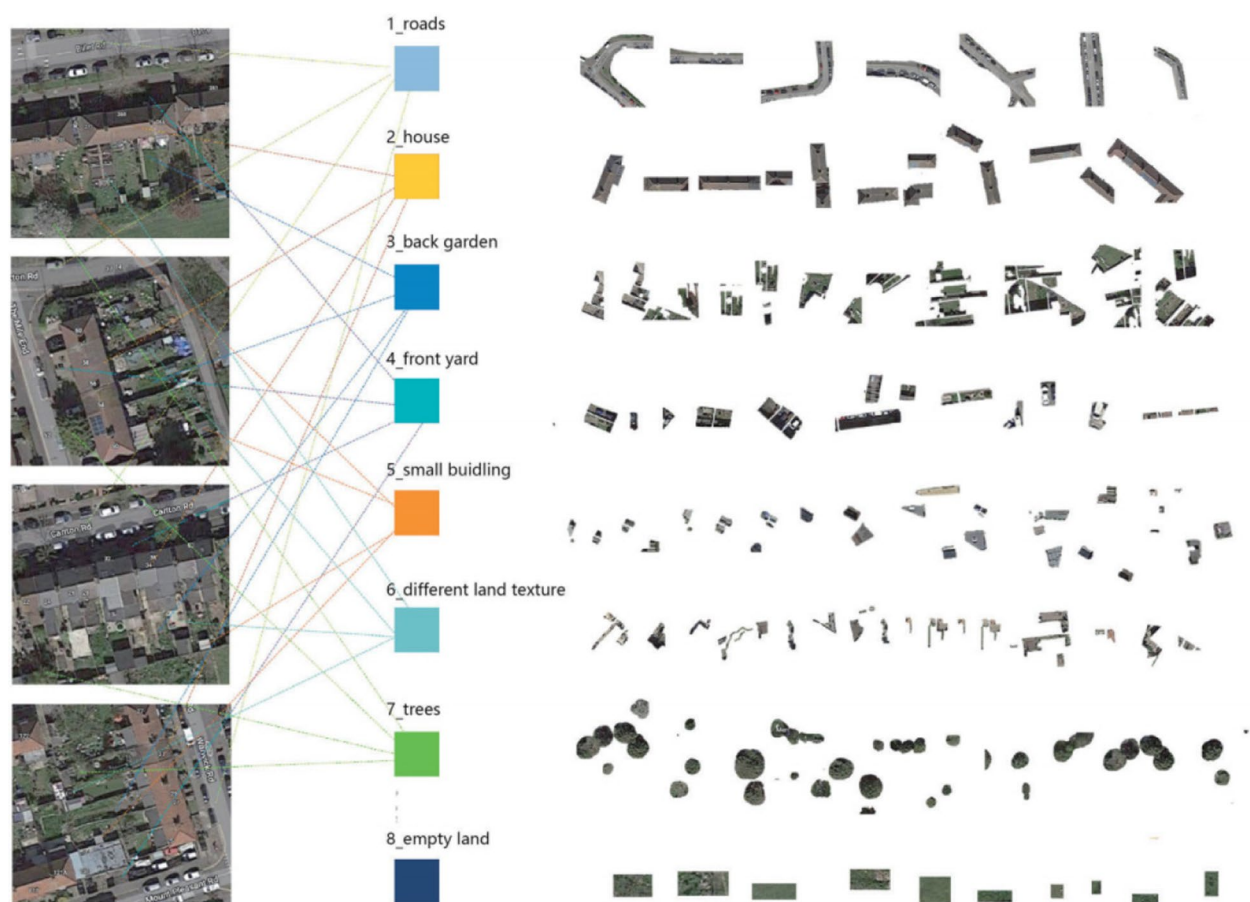


Fig. 9 The list of segmented features for pix2pix training

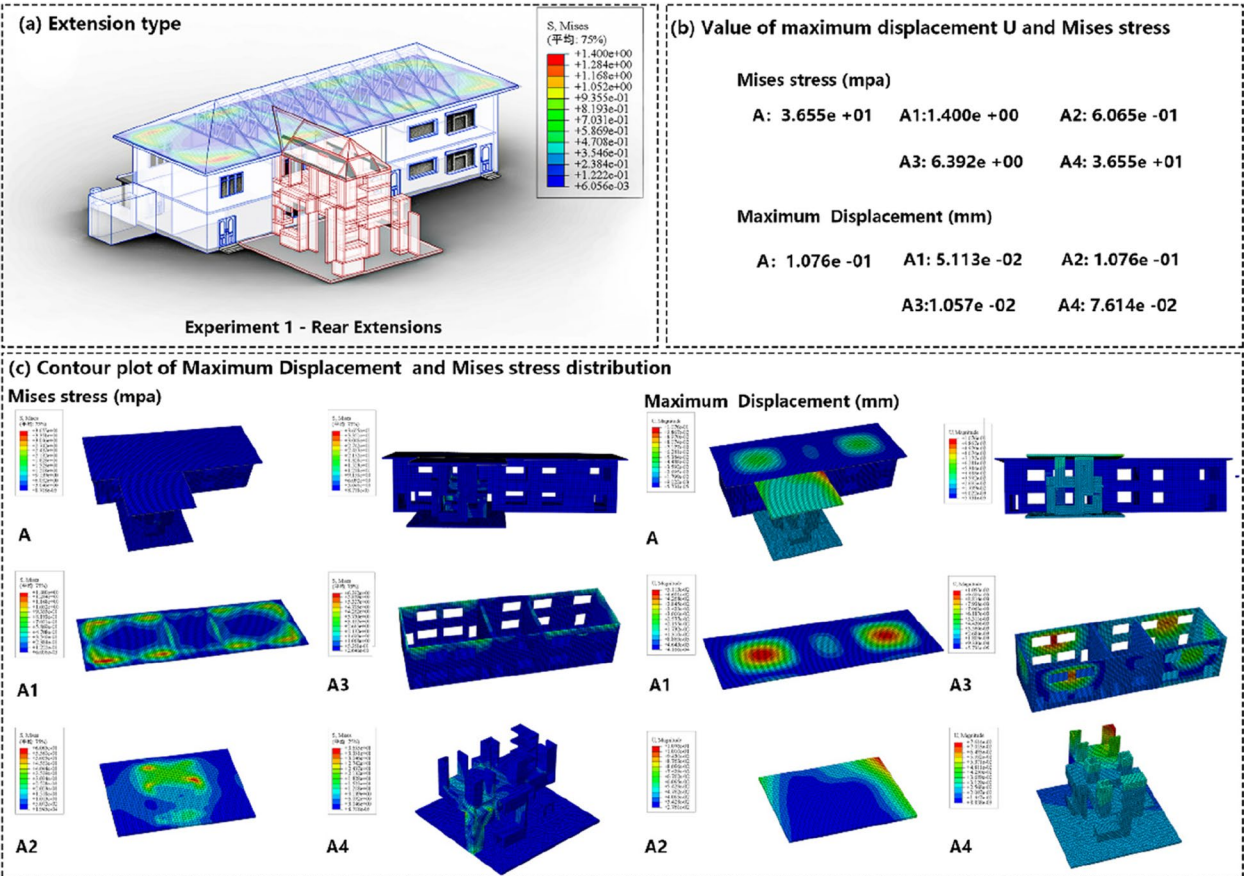


Fig. 10 Experiment 1 (a) extension type 1 (b) value of maximum displacement and mises stress (c) contour plot of maximum displacement and mises stress distribution

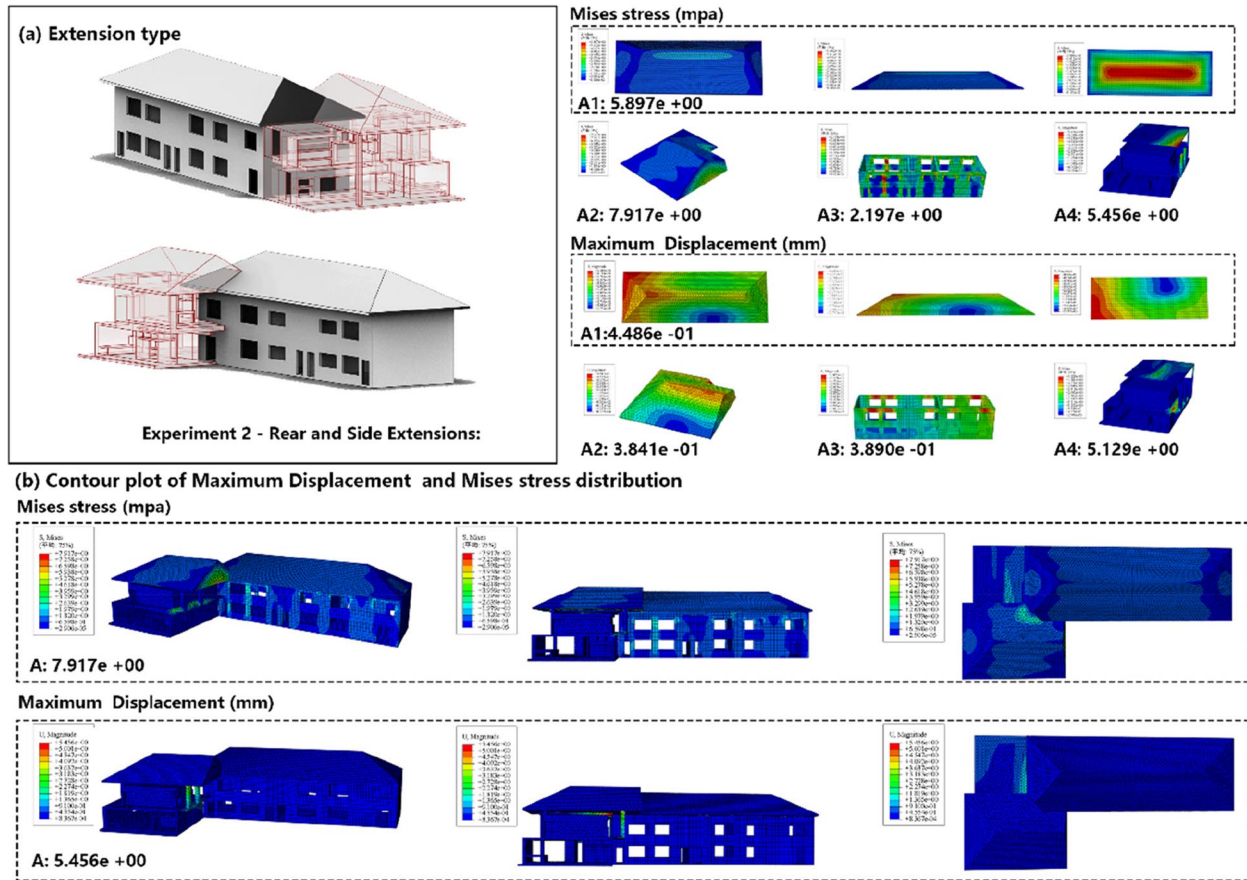


Fig. 11 Experiment 2 (a) extension type 2, (b) contour plot of maximum displacement and mises stress distribution (c) value of maximum displacement and mises stress

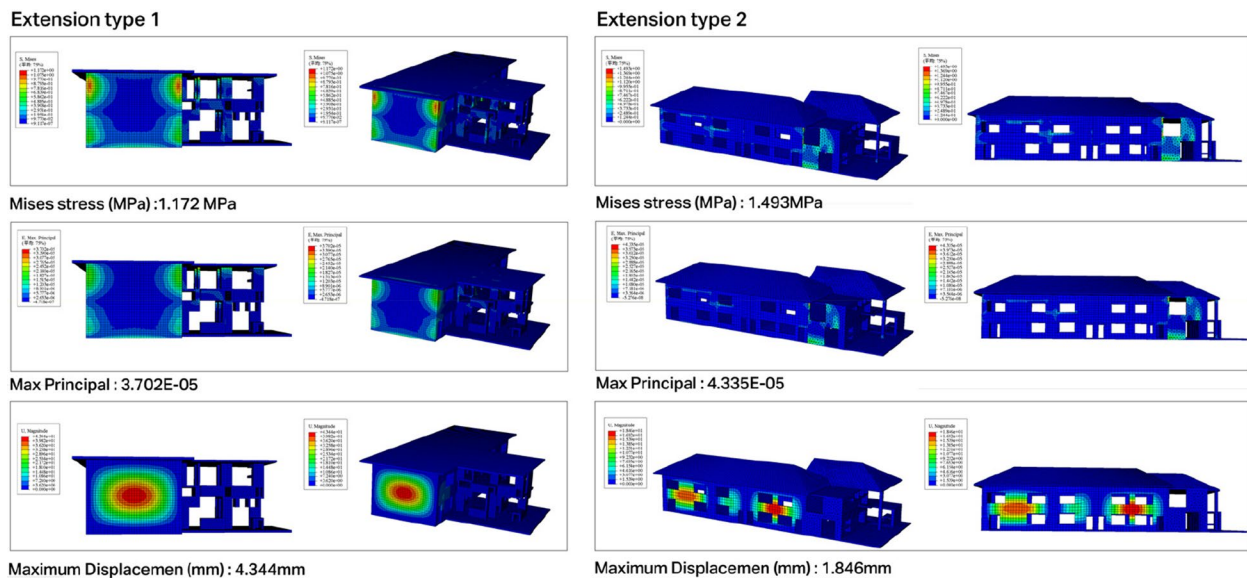


Fig. 12 Experiment 3 Wind load of two different house extension

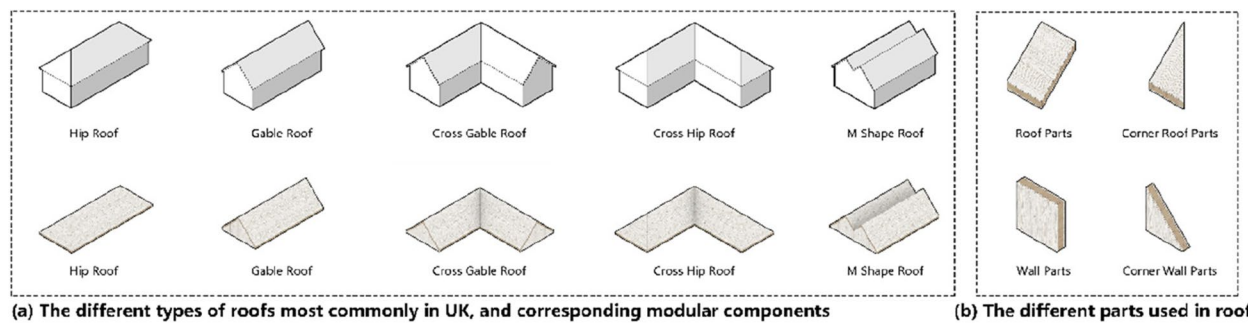


Fig. 13 (a) The different types of roofs most commonly in the UK and corresponding modular components (b) The different parts used in the roof

Acknowledgements

The authors would like to express our heartfelt gratitude to our colleagues, Mengzhen Guo and Yusong Hu, for their contributions to this project's initial ideas. We also thank Boyuan Yu and Jianing Luo for their assistance with background research. Additionally, we sincerely thank Professor Gilles Retsin, Manuel Jimenez Garcia, and Mollie Claypool of the Bartlett School of Architecture, UCL, for their invaluable advice and guidance throughout the preliminary research process.

Authors' contributions

Conceptualization: Yangzhi Li and Jingwei Li; Methodology: Yangzhi Li and Jingwei Li; Formal analysis and investigation: Yangzhi Li and Qiwei Song; Writing—original draft preparation: Yangzhi Li; Writing—review and editing: Yangzhi Li and Qiwei Song; Supervision: Yangzhi Li and Qiwei Song. All authors read and approved the final manuscript.

Funding

No funding support.

Data availability

The data and materials in this work are available upon request. Codes and training data are available at: <https://github.com/liyangzhi-svg/Extending-Roof-Ridge-Lines-for-Sustainable-Housing-Extension.git>.

Declarations

Competing interests

The authors declare no conflicts of interest.

Received: 4 June 2025 Revised: 9 August 2025 Accepted: 28 August 2025
Published online: 10 October 2025

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